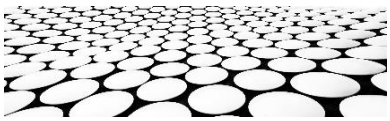
CSI4142 Fundamentals of Data Science



ASSIGNMENT 2

Data cleaning

GOALS

The overall goal of this assignment is to become familiar with Data Cleaning, more specifically with duplicate detection, data validation processes, as well as imputation for missing values.

At the end of this assignment, you will have:

- Reviewed your Python skills, as the project MUST be done in Python
- Explored Kaggle, an amazing resource of challenges and datasets
- Programmed a "validity checker" that can run on a dataset to find possibly invalid data
- Programmed and evaluated imputation methods for missing data
- Documented everything, in Jupyter Notebooks, in a way to make your analysis understandable and reproducible.

PLEASE NOTE: During the lectures, we have also covered outlier detection in our exploration of data cleaning. This topic is not included in the current assignment. We will revisit outliers in a later assignment.

SUBMISSION INFORMATION

- Deadline:
  - Submission of your notebook: **Tuesday, March 3rd, midnight**
  - Grace period (no penalty) is until 12:30… after that it's 20% per day off.
- Groups:
  - You are expected to form groups of 2 and do a single submission per group.
  - You need to register your group in Brightspace.
  - It is not possible to work alone, and to enforce such policy, a penalty of 20% will be imposed on solo groups. Make sure you find a partner through the Group Search sheet on Brightspace. Or even use the Forum if you prefer.
- Where to submit:
  - Your submission must be done in Brightspace in Assignment section (Assignment 2)
- Submission format:
  - <u>Only Jupyter Notebook files will be accepted to show your programs and results</u>
  - For the datasets used in your analysis, your code MUST contain a link to the dataset so the TAS do NOT need to download any data.
- Demo/Presentation (**mandatory**):
  - You must select a time (a set of timeslots will be provided) to meet with one of the TAs to present your work.
  - Both students must be present during the presentation. If one student is missing, that student will have a 20% penalty on the entire assignment. The student who missed will not be allowed to reschedule as the demo is already done.
  - A group that has a valid reason can contact the TA to reschedule without penalty, but it must be done at least 24 hours before your time slot, and there must still be available timeslots. A group cannot reschedule twice.
  - If you miss your time slot, it is 20% penalty on the entire assignment. To not have zero, you should reschedule within 24 hours of your missed slot, and while there are still available timeslots.
  - Any no-show will be given zero on the assignment. **Even if the demo part counts for 12 marks out of 60, not doing the demo results in zero for the entire assignment.**

PLEASE NOTE:
Even if you do a demo, the correctors might wish to run your code at a later time. If the corrector cannot run your code, the mark will be zero for the entire assignment. It is your responsibility to test that the code cells in your Notebook are executable. The <u>most frequent problem</u> is that the dataset is not readable (file not found). Please make sure the dataset is either read directly from a public place (a dataset repository) or read from one of your shared repository (accessible to anyone) so the TA can run your notebook without any data download.

TUTORIALS/TECHNOLOGIES

A series of links to different videos (Python, Jupyter, Colab, etc) was provided for Assignment 1. Please take a look at these links if you are still unsure about various aspects of programming for your assignment 2.

INSTRUCTIONS

1. Choose two datasets

The Kaggle site is a very good site to explore as it contains datasets for many tasks in data science and AI. There are also other sites for datasets if you wish to further explore. You must choose 2 datasets, and *they cannot be the same as the ones from your Assignment 1.*

Dataset 1 – Validity checking

The first dataset will serve as an example for your clean data checker (described in Section 2).

For **Dataset 1**, you are free to choose any dataset that you want. Make sure it has multiple attributes on which you will be able to artificially introduce noise and then use algorithms to detect it. The noise can be introduced manually or with simple algorithms.

Dataset 2 – Imputation

The second dataset will serve as an example for the imputation methods (described in Section 3).

For **Dataset 2**, you can choose any dataset that you want (not the same as Dataset1). Make sure it has multiple attributes on which you will be able to artificially remove data then use algorithms to do imputation. The removal can be done manually or with simple algorithms.

ATTENTION: Both Dataset 1 and Dataset 2 must be real datasets. Synthetic datasets are NOT accepted.

Please note: During the semester, for your assignments, you will need to choose various datasets for different explorations so you are exposed to data from different domains. You are NOT allowed to reuse a dataset from one assignment to another. So, if you find a few datasets that you think are interesting, save some for your future assignments!

2. Validity checker

In this part of the assignment, you are asked to do a "Validity Checker". In class, we have seen 9 potential validity problems, of which 8 are listed below (I removed "code errors" as it relies on an external resource). Your checker will need to perform 8 validity tests:

1) Data Type errors
2) Range errors
3) Format errors
4) Consistency errors
5) Uniqueness errors
6) Presence errors
7) Length errors
8) Look-up errors

You are asked to program a "validity checker" which would be able to detect any of the 8 types of invalid data. To be able to evaluate the usefulness of your validity checker, you will intentionally introduce errors in the dataset and then use the checker to find it. Think of modifying the data to introduce about 5% of error. Make sure you keep the original dataset before you modify it.

For each type of validity check, you should have 4 cells in your Notebook.

- Cell 1: Type of error: name and description
- Cell 2: How you introduced such error (which column? How?)
- Cell 3: Code that will automatically find the error.
- Cell 4: Qualitative results in the form of explaining sentences + examples.

I show an example on next page.

Cell 1 – Test name + description

Range check:

In this test, we will verify the range of a numerical value. The range is the minimum and maximum values that an attribute can have.

Cell 2 – How you introduced such error

Code that shows that 5% of the rows, the column “age” is assigned a random number between -20 and 140.

Cell 3 – Checker code

Code that shows the range detection (e.g. valid age from 0 to 120).

Cell 4 – Report of findings

Showing a few examples of data points where the issue is solved. You are not asked for a quantitative evaluation at this time, so only putting 3 examples of before/after is fine.

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3. Missing data and Imputation

In this part of the assignment, you are asked to devise an experiment in which you perform 3 imputation tests, each test is as follow:

a) Choose an attribute on which to test your imputation method
b) Simulate missing values by removing some values on that attribute in the dataset
c) Program an imputation approach for replacing the missing values
d) Evaluate to what extent your approach is finding the missing values (we will discuss quantitative evaluation approaches in class)

Your 3 tests must be on different attributes. For each test, for (b) you should mention if you are simulating MCAR, MAR or MNAR for your removal.

For (c), you must choose among the list below an imputation method that seems appropriate. All 3 tests must use a different imputation method (you cannot consider mean and median as 2 different methods for example as they both fall under Default Value imputation below). You must program 3 methods. I suggest to program one univariate (e.g. median), one bivariate (correlation or regression) and one multivariate (e.g. similarity-based).

Choice of imputation methods:

1. Default value imputation
2. Random Sample imputation
3. Time-based imputation
4. Correlation imputation
5. Regression imputation
6. Predictive imputation
7. Similarity-based imputation

Similarly to your work on Validity Check, you should have 4 cells in your Notebook for each of the 3 imputation methods chosen. Cell 4 though should contain an evaluation of the method.

- Cell 1: Type of imputation: name and description
- Cell 2: How you introduced missing data (which column? How? Are you simulating MCAR, MAR or MNAR)
- Cell 3: Code that will automatically find the missing data and replace it.
- Cell 4: Quantitative evaluation of the method.

☆☆☆
EVALUATION (60 points)

Jupyter Notebook (48 points)

- Overall effort in the report (4 points)
  - Provided all required sections clearly identified
  - Good cell separation (text, code, results, etc)
  - Tests on various examples easy to perform by the corrector
  - Report detailed enough for reproducibility
- Heading (-3 if missing)
  - Group number + student names + student numbers
  - Work split (provide information about who did what for the assignment, explain the work split between the 2 students as to know who was responsible for what)
- Datasets descriptions (4 points)
  - Dataset name, author, purpose (what was it made for, NOT synthetic)
  - Shape: how many rows and columns
  - A list of the features + descriptions (what do they represent and are they categorical or numerical)
- Validity checker (26 points – 8 tests * 3 points + 2 for overall structure)
  - All 8 validity tests are running
  - Four cells for each test are clear (validity type, noise introduction, validity checking, qualitative evaluation)
- Imputation (14 points – 3 test * 4 points + 2 for overall)
  - Choice of 3 approaches
  - Programming of random replacement with missing values
  - Programming of each approach
  - Quantitative Evaluation of the 3 approaches
- References (-3 if missing)
  - For any part of your code taken from a web site (even a tutorial site or stackoverflow), you must provide the reference to it.
  - If part of your code was generated using Generative AI, please list the tool as well as the queries performed for each code snippet used.
  - Any theory/algorithms found in books, slides, tutorials that you used should be referenced.

Presentation/Demo (12 points)

- Understanding of the work
  - Students show a good understanding of the work they have done.
- Clarity of explanations
  - Students clearly express themselves using appropriate terms to describe what they have done.

QUESTIONS

- You can ask your questions within the assignment topic of the discussion forum on Brightspace.
- To make sure you get answers on the forum on a regular basis, the 3 TAs will share the days to answer your question. Notice that they are NOT available on weekends.

	Forum Monitoring						
	Monday	**Tuesday**	**Wednesday**	**Thursday**	**Friday**	**Saturday**	**Sunday**
Morning (before noon)	Svetlana	Ángel		Ángel	Svetlana		
Evening (before 9pm)	Shavbo	Shavbo	Svetlana	Shavbo	Ángel		

If you need to reach the TAs by email, you should use the following emails:

- Shavbo: ssale038@uottawa.ca
- Svetlana: sesina@uottawa.ca
- Ángel : avale073@uottawa.ca